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Assignment-6

Imbalance Learning(2000095)

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Credit Card Fraud Detection: Addressing Class Imbalance

In this project, we aimed to develop a predictive model that can accurately detect fraudulent transactions while taking into account the significant class imbalance present in the data. Our dataset consisted of 284,807 transactions, with only 492 instances being fraudulent (Class 1). This extreme imbalance posed a challenge for traditional machine learning algorithms, which tend to become biased towards the majority class.

Data Exploration and Preprocessing:

Before diving into model development, we explored the dataset to understand its characteristics. We found that the data was complete, requiring no imputation. We then standardized two numerical features, Time and Amount, using StandardScaler to normalize the data. Finally, we split the data into training and testing sets (80/20) while preserving the minority class ratio.

Class Imbalance Handling Challenges:

The primary challenge posed by this dataset is that standard machine learning algorithms tend to ignore the minority class (fraudulent transactions). A model could simply predict "non-fraudulent" for every transaction, achieving over 99% accuracy but failing to detect actual fraud. To address this issue, we proposed and implemented several strategies:

Resampling Techniques: We used Synthetic Minority Over-sampling Technique (SMOTE) to generate synthetic data points for the minority class based on the feature space of existing fraud cases.

Ensemble Methods: We employed Random Forests, which aggregate multiple decision trees to isolate minority class regions without severely overfitting.

Model Development and Evaluation:

We selected Logistic Regression and Random Forest as our machine learning algorithms. We trained and evaluated the models using precision, recall, and F1-score to account for the imbalance in the dataset. Our findings showed that:

Model 1: Logistic Regression (No Balancing): The baseline model achieved a decent precision but missed many actual fraud cases.

Model 2: Logistic Regression (with SMOTE): Applying SMOTE improved recall but introduced many false positives.

Model 3: Random Forest (with SMOTE): This combination yielded the best results, detecting fraudulent transactions while keeping false positives in check.

Discussion and Conclusion:

In summary, our approach demonstrated that addressing class imbalance is crucial. While relying on resampling alone may not be sufficient, combining it with an ensemble method like Random Forest can yield impressive results. Our findings highlight the importance of carefully considering class imbalance when developing predictive models for detecting credit card fraud.

Future directions for improvement include exploring cost-sensitive learning algorithms, undersampling the majority class, or utilizing advanced deep learning techniques like Autoencoders for anomaly detection.